

Genetic Algorithm

GA has majorly three applications, namely, intelligent search, optimization and machine learning.

Intelligent search: The search problems in CS are of deterministic in nature, i.e., the order of visiting the elements of the search space are known. ▽

Ex- Depth first / Breadth first search-

However the search problems in AI are non-deterministic and thus visiting the elements of the search space (nodes of the tree) are completely dependent of data set.

Optimization: It is the minimization / maximization of an objective function. The objective function represents the problem space for a data set. — Fitness Function

Machine learning: These are computer programs to corresponding to some task and with respect to some experience. Thus the performance of the computer is measured (Performance measure) and improves its experience. In every simple words, Machine learning is giving computer human intelligence.

GA is used along with neural nets and fuzzy logic for solving complex problems. These together are often referred to by a generic name Soft computing.

GA operates through simple cycle of stages:

- i) Creation of population of strings
- ii) Evaluation of each string — Fitness f(x)
- iii) Selection of best string
- iv) Genetic manipulation to create new population of strings

Chromosomes
↓
Possible soln.

Offspring
New generation

Population
chromosomes

Decoded String

Fitness
Evolution

Genetic
operators

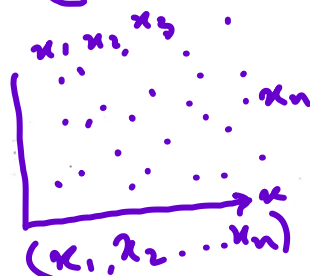
Parents

Selection
Mating Pool

Manipulation

Reproduction

$$\begin{cases} f(x) = 3x^2 - 2x \\ x \rightarrow f(x)_{max} \\ 0-50 \end{cases}$$



✓ 9, 31, 26 chromosomes

Initial population — representatives of the potential solution is created to initiate the search process

Chromosomes — The elements of the population encoded into bit string. → 1 — 00001 → chrom.

Fitness — The performance of the string — evaluated with help of ~~few~~ some function — fitness function

✓ Thus depending on the fitness of the chromosomes, they are selected ~~as~~ for genetic manipulation process.

✓ It is to be noted that the selection process is mainly responsible for assuring survival of best-fit individuals.

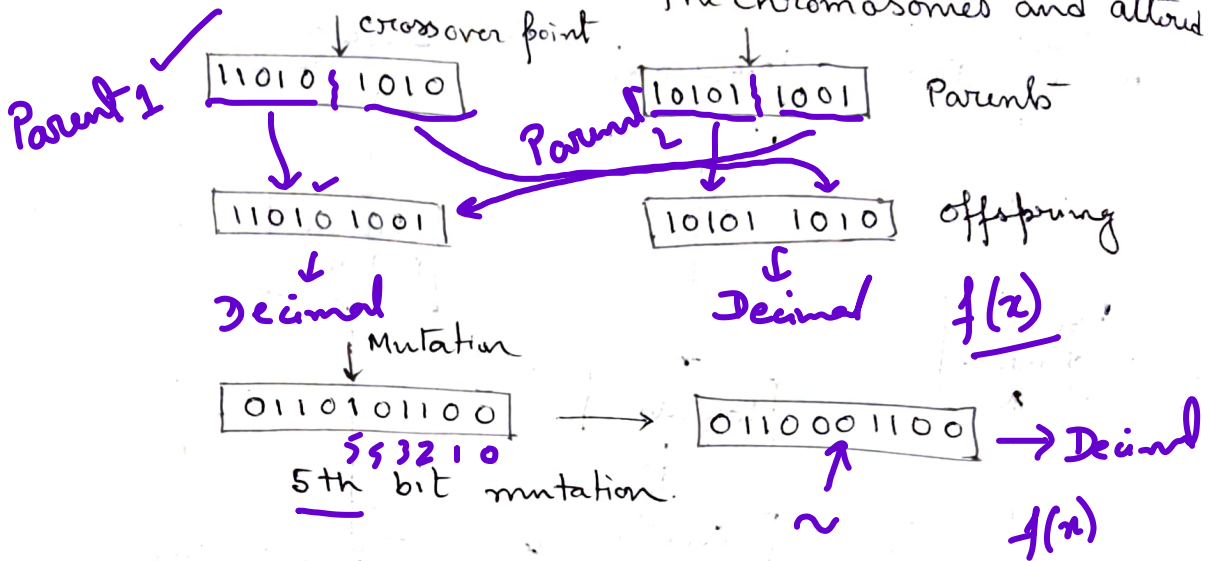
Genetic Manipulation

Cross over

Recombination of bits of each two selected strings

Mutation

One or more bits are randomly selected of the chromosomes and altered



Ex. Find solution by GA cycle for $f(x) = x^2$ in the interval $0 \leq x \leq 31$

Step I: Initial population

String No.

Chromosome

1	01101	13
2	11000	24
3	01000	08
4	10011	19

Randomly

Step II: Form fitness function

a) $f(x) = \text{fitness of } x$
 $= x^2$

fitness: $f(x)_1 = 169$ $f(x)_2 = 576$ $f(x)_3 = 64$ $f(x)_4 = 361$

b) Normalization of strength fitness

$$p_i = \frac{f(x)_i}{\sum_{i=1}^n f(x)_i}$$

$p_1 = \frac{1000}{1000+492+55+300} = 0.144$
 $p_2 = \frac{492}{1000+492+55+300} = 0.492$ ✓
 $p_3 = \frac{55}{1000+492+55+300} = 0.055$ ✓
 $p_4 = \frac{300}{1000+492+55+300} = 0.309$

The selection of chromosomes are done on the basis of crossover probability. The chromosomes having fitness values less than probability are rejected as bad genes.

Cumulative Probability
 $q_1 = p_1$
 $q_2 = q_1 + p_2 = \dots$

Step III : Manipulation

The strings having large fitness value is identified as the better string.

Here 2nd string is having the highest value, thus selected twice, 4th string & 1st strings are selected once respectively.

* These selection can also be done depending on Crossover probability.

String No.	Mating Pool	Mating String	Cross over Point	New Population
1	<u>01101</u>	<u>11000</u>	2	<u>01100</u>
2	11000	01101	1 ✓	<u>11001</u>
2	<u>11000</u>	<u>10011</u>	4 ✓	<u>11011</u>
4	10011	<u>11000</u>	2	<u>10000</u>
Parents				Offsprings

✓ The new solutions are feasible & global/local optimal solution.

⊛ A random number is generated, say if the random number is 3 then the crossover point is selected in between three and four.

Schema

A chromosome with fixed bits (1/0) at designed points.

eg. $\checkmark$ $\underline{? ? 1 ? 0 1 0 ? ?}$
8 7 6 5 4 3 2 1 0

If fixed bit is altered then it is not a schema

The fixed bits are 2, 3, 4, 6

The other bits are known as don't care bits, represented by '?'

Since four positions ^{matter} for this schema, we say the schema contains four ~~bits~~ genes.

Order of Schema

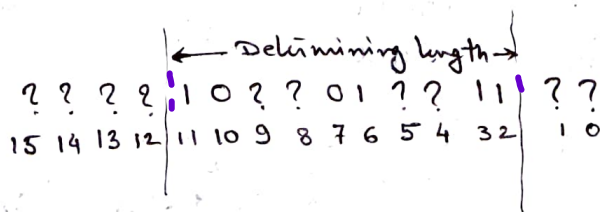
It is denoted by $O(H)$, where H is the schema

The number of the fixed bits in schema is the order

$$O(H) = 4$$

Defining Length

It is the difference between the largest and smallest fixed bit position.



$$d(H) = 11 - 2 = 9$$

sol.
 $x = 15, 50, 50, 32, \dots$
 $x = 21, 26, \dots$
 $f(x) \uparrow$

The Fundamental theorem of Genetic Algo (Schema Theorem)

A schema with ~~an~~ above ~~the~~ average fitness will have an exponential rise and thus would finally occupy almost the entire population pool.

This observation was made by Prof. Holland.

Let, # Schema $H = 11??00$

	x_1		x_2	
=	11	00	00	•
	11	01	00	
	11	10	00	
	11	11	00	

fitness

For some chromosomes the fitness values are different. Thus average value of the schema pool can be calculated. Schema value may be more than average value.

f_H = average fitness of schema H

f_{avg} = average fitness of population N

$f_H > f_{avg}$ means schema H will increase exponentially until entire population is occupied by schema H .

Let,

N - population size

$m_H(t)$ - samples of schema ' H ' at generation ' t '

Among the selection strategies, the most common is the proportional selection. In proportional selection the number of copies of chromosomes selected for mating is proportional to their respective fitness values. Thus, following the principles of proportional selection, a string ~~selection~~ ' i ' is selected with probability

$$f_i / \sum_{i=1}^n f_i$$

(*) The probability that the chromosome will not be damaged
 $= 1 - p_c \frac{d(H)}{(L-1)}$

where f_i is the fitness of string 'i'.

Now the probability that in a single selection a sample of schema H is chosen is described by

$$f_H = \frac{\sum_{i=1}^{m_H} f_i}{m_H} \quad \frac{\sum_{i=1}^{m_H(t)} f_i}{\sum_{i=1}^n f_i} = m_H(t) f_H / \sum_{i=1}^n f_i$$

$$\sum_{i=1}^{m_H} f_i = f_H \cdot m_H$$

f_H is the average fitness of the $m_H(t)$ samples of H

Thus, in a total of N selections with replacement, the expected number of samples of schema H in the next generation is given by

$$m_H(t+1) = N m_H(t) f_H / \sum_{i=1}^n f_i$$

$$= \cancel{m_H(t) / f_{avg}} \cdot m_H(t) f_H / f_{avg}$$

where $f_{avg} = \sum_{i=1}^n f_i / N$ (*) if $f_H > f_{avg}$ then $m_H(t+1) > m_H(t)$

It indicates that the GA allocates over an increasing number of trials to an above average schema.

Effect of change of Schema due to cross over;

$H = ??0??(?)1$
 $X = 11010(0)1$
 ← crossing site

if offspring

1101000
 0010001

Thus, the probability of the schema H dying out is given by

$$p_d = d(H) / (L-1)$$

Fixed bits have changed

Thus schema H does not exist

(*) Probability of cross over site will be in the region of defining length = $\frac{d(H)}{(L-1)}$

Selection of Best Chromosomes

According to ~~the~~ Darwin's principle the best chromosomes survive to create new offspring. There are many methods -

✓ i) Roulette wheel selection



✓ ii) Rank selection

✓ iii) Tournament selection

~~iii~~

<i> Parents are selected according to their fitness. The better chromosomes are, the more chance to get selected

All the chromosomes in the population are placed on roulette wheel, and every chromosome has its place according to its fitness.

<ii> The previous selection has a problem if the fitness differ very much. For example, if the best fitness of chromosome is 90% on the wheel then the other chromosomes have a very few chance to get selected.

Rank selection first ranks the population and then every chromosome receives fitness from this ranking. The worst will have Rank 1 and the best fitness will have N (number of chromosomes in population).

✓ Exp. value $(i, t) = \text{Min} + (\text{Max} - \text{min}) \frac{\text{rank}(i, t) - 1}{N - 1}$

* If p.c. is the crossover probability then the probability that the chromosome of schema 'H' will be damaged due to crossover = $\underline{p_c} \times \frac{d(H)}{(L-1)}$ p_c - crossover probability